Sunny Liang Tan (谭靓)

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References

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Education

Oct 2021 -	**University of California, Los Angeles (UCLA)** Ph.D. in Economics Expected Graduation: Jun 2027	Los Angeles, CA
Sep 2024 - May 2025	**Massachusetts Institute of Technology (MIT)** Visiting Scholar	Cambridge, MA
Sep 2019 - May 2021	**Duke University** M.A. in Economics	Durham, NC
Sep 2015 - May 2019	**The University of Hong Kong** B.A. in Economics & Finance; Minor in Mathematics	Hong Kong

Research Fields

Primary: Macroeconomics, International Economics
Secondary: Development Economics, Industrial Organization
Research Topics and Methods: Industrial Policy, Production Networks, Microeconometric Identification and Macroeconomic Aggregation, GE Sufficient Statistics, Chinese Economy

Job Market Paper

Government Venture Capital, Production Networks, and Aggregate Productivity in China

Abstract: Industrial policy can reshape production well beyond the firms it directly supports, and assessing its aggregate productivity implications therefore requires measuring how supply-chain spillovers amplify or counteract recipient gains. This paper provides the first joint analysis of these direct, upstream, and downstream contributions in the setting of government venture capital in China. Linking investment records to firm accounts, patents, and supplier–buyer transactions, we identify expansion among financed firms and their existing trading partners. Suppliers expand sales and capital, while buyers expand sales and payroll. We combine these responses with baseline allocation statistics in a production-network framework that values both the resources absorbed by expanding firms and those released elsewhere. In a broad-population calibration with fixed industry factor supplies, supplier and buyer spillovers amplify the recipient contribution, with the buyer contribution accounting for most of the total. Positive labor-reallocation contributions outweigh the negative capital-reallocation contributions at trading partners. The analysis shows that production networks can substantially enlarge the productivity contribution suggested by recipient outcomes alone, highlighting their importance for evaluating industrial policy.

Working Paper

Exchange-Rate Movements and U.S. Corporations with Ivo Welch and Yuqing Zhou

Abstract: This paper revisits the exchange-rate exposure puzzle: why prior research finds weak relationships between currency movements and U.S. firms' stock returns despite substantial international trade. Linking transaction-level U.S. Census export records to financial statements and stock returns for publicly traded U.S. exporters from 1992 to 2016, we construct firm-specific exchange-rate exposure measures that reflect each firm's export destinations. Dollar depreciation is associated with economically significant increases in exports, sales, profitability, and stock returns, with export, sales, and stock-return sensitivities generally larger among larger and more export-oriented firms. In asset- or export-weighted specifications, a 1% depreciation is associated with approximately 0.4-0.6% higher exports and 0.6-0.8% higher sales. The stronger sales response is consistent with currency movements affecting domestic competitiveness as well as foreign demand, while the profitability results indicate that financial and operational hedging do not fully offset exposure. Real operating sensitivities increase over time. Stock prices respond contemporaneously, with no detectable delayed response around subsequent earnings announcements. By measuring exposure using firms' actual trading relationships and tracing its implications across operating outcomes and equity valuations, the paper contributes new evidence to the exchange-rate exposure literature and shows that substantial corporate currency exposure can coexist with timely incorporation of information into stock prices.

Work in Progress

Industrial Policy and Scale Complementarities in EV Battery Supply Chains with Junjie Xia and Binkai Chen

Abstract: This project studies whether scale economies at successive production stages amplify industrial subsidies through mutually reinforcing demand and cost linkages in EV battery supply chains. Support for battery producers can increase demand for materials, allowing suppliers to lower unit costs; when passed downstream, these savings can induce further expansion. Using linked income-tax, VAT, and firm-to-firm transaction data, the analysis traces policy exposure through existing supplier relationships and examines suppliers' scale, costs, investment, and sales to other buyers. It distinguishes scale economies from changes in demand, markups, product composition, and supplier selection, and will discipline a production-network model to compare upstream, downstream, and joint support under a common fiscal budget.

Industrial Policy and Induced Innovation in Semiconductor Supply Chains with Mervyn Ho, Junjie Xia, and Binkai Chen

Abstract: This project studies whether support for downstream chipmakers induces technological upgrading among semiconductor equipment suppliers. More advanced chip production raises equipment requirements, potentially making costly supplier innovation worthwhile, while repeated testing and customer feedback may generate learning that benefits subsequent buyers. Combining government venture capital investments, administrative firm data, supplier–buyer transactions, and fabrication-capacity and process-technology data, the analysis distinguishes customer capacity expansion from technological advances and examines how exposure duration and intensity shape supplier upgrading. It investigates whether new capabilities extend to other buyers and separates upgrading within incumbent suppliers from substitution toward suppliers that already possess advanced capabilities, tracing transferable learning beyond directly financed firms.

Multinational Investment and Regional Growth in Mexico with Martin Tobal

Abstract: How does foreign investment translate into productive capabilities in a host economy? This paper develops an empirical framework for studying Chinese multinational investment in Mexico following the U.S.-China trade war. The central hypothesis is that foreign entry can generate persistent regional gains through worker learning, improved labor-market matching, and spillovers to domestic firms. These mechanisms must be distinguished from the direct increase in Mexican export demand caused by U.S. tariffs on Chinese goods. We propose linking employer-employee records, manufacturing establishment surveys, foreign-ownership information, and trade data to estimate regional responses and trace workers across foreign and domestic employers. The regional design combines predetermined employment shares with sectoral Chinese investment in Vietnam as a candidate instrument for Mexican investment. Worker-mobility analysis would separate employer wage premia from earnings gains that persist after foreign-firm experience. A quantitative spatial framework would connect these responses to migration, local prices, and welfare.

Multinational Networks and U.S.-China Semiconductor Trade

Abstract: Does geographic diversification of U.S. semiconductor imports reduce exposure to China, or reorganize trade within multinational firms? This project proposes to link LFTTD semiconductor transactions with historical ownership data from Orbis and supplementary shipment records. It will trace imports supplied by Chinese-controlled affiliates in third countries and examine whether U.S. importers switch sourcing locations while retaining the same corporate suppliers following tariff changes. The analysis will quantify ownership-based exposure to China, distinguishing corporate control from Chinese input content, and assess how multinational networks shape the interpretation of semiconductor trade decoupling.

Teaching Experience

University of California, Los Angeles (UCLA)
Teaching Assistant, Principles of Economics (Winter 2024, Spring 2026)
Teaching Assistant, Microeconomic Theory (ECON 101; Fall 2022, Winter 2023, Fall 2023)
Teaching Assistant, Microeconomic Theory (ECON 11; Winter 2026)
Teaching Assistant, Ph.D. International Macroeconomics (Spring 2024)
Teaching Assistant, Competitive Strategy (elective; Spring 2023)
Duke University
Teaching Assistant, Ph.D. Econometrics II (Fall 2020)
Teaching Assistant, Econometrics and Data Science (Spring 2021)

Fellowships and Awards

2021–2027	UCLA Graduate Fellowship
2020	M.A. Merit Scholar Award, Duke University
2018	HKU Centennial Scholarship
2015–2016, 2017–2018	Dean's Honors List
2016	Hong Kong SAR Government Reaching Out Award

Presentations and Professional Service

- Presentation: South Cal International Macro Finance Conference, 2026.
- Presentation: West Coast Micro-to-Macro Development Workshop, 2026.
- Presentation: MIT Trade Lunch, 2025.
- Organizing Committee, Southern California Graduate Conference in Applied Economics, 2023.
- Initiator, UCLA Women in Economics, 2023-2025.

Research Experience

Jan 2023 - Feb 2023	**University of California, Los Angeles (UCLA)** Research Assistant to Jon Vogel Theoretical Analysis of Labor-Market Equilibrium	Los Angeles, CA
Sep 2021 - Nov 2021	**Duke University** Research Assistant to David Berger Literature Reviews and Data Construction on Wages and Taxes	Durham, NC
Jan 2021 - Jun 2021	**Duke University** Research Assistant to Matthew Masten Econometric Proof Summaries and Literature Reviews	Durham, NC
Nov 2018 - Jul 2022	**Massachusetts Institute of Technology (MIT)** Research Assistant to David Autor Occupational Data Digitalization, Classification, and Econometric Analysis	Cambridge, MA

Other Research Experience

Jan 2026 - May 2026	**SoftBank China Venture Capital** Data Collaboration China Artificial Intelligence and Energy	Hybrid
2023	**Price Theory Summer Camp**	Chicago
Jun 2017 - Jul 2017	**Bank of China (Hong Kong)** Summer Analyst Bond Connect Due Diligence / Regulatory Materials	Hong Kong
Dec 2016 - Jan 2017	**Muxin Asset Management (Shanghai)** Quantitative Research Intern A-Share Alpha Strategies / Foreign Exchange Research	Shanghai

Languages

Mandarin (native); English (fluent); Cantonese and Spanish (conversational)

Last updated: November 2026